Jiakun Fan

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Research focus: MLSys, Efficient LLM Inference, RSI Infrastructure

Education

Virginia Tech Sep. 2024 – Present
Ph.D. in Computer Science; expected graduation: 2028 *GPA: 4.0/4.0*

The Chinese University of Hong Kong, Shenzhen Sep. 2020 – Jun. 2024
B.Eng. in Electrical and Computer Engineering

Experience

ByteDance Summer & Fall 2026
Research Scientist Intern, ML Systems *San Jose, CA*

- Research on efficient LLM inference, including diffusion-based speculative decoding and training–inference co-design.

Virginia Tech – PEARL Lab
Former Graduate Research Assistant *Blacksburg, VA*

- Research on memory-efficient LLM serving, heterogeneous CPU–GPU execution, and distributed speculative inference.

Selected Research Projects

Long-Horizon Autonomous Research Infrastructure Fall 2026

- Built infrastructure for reliable long-horizon AI research loops, coordinating agent-driven code changes, isolated GPU experiments, candidate evaluation, and persistent experiment lineage.
- Developed the execution protocol and recovery mechanisms for concurrent workers, atomic submissions, result delivery, and process/storage failures; validated key invariants with fault-injection tests and TLA+ models.

DScout: Diffusion-Drafted Speculative Decoding Summer 2026

- Developed prefix-aware diffusion-drafter training through mixed horizons, intra-block unmasking, and acceptance-weighted loss.
- Optimized inference configurations using acceptance–latency analysis and block-size selection without retraining; achieved 4.14× mean speedup over autoregressive Qwen3-8B decoding in single-request, greedy benchmarks.

Efficient KV Cache Management for Agentic LLM Serving Spring 2026

- Investigated KV cache management for long-context, multi-turn LLM workloads, focusing on memory pressure and serving efficiency.
- Implemented and evaluated serving optimizations in SGLang, studying their impact on latency and throughput across agent workloads.

dLLM-Serve: Memory-Efficient Diffusion LLM Serving Spring 2025

- Built a diffusion LLM serving system addressing transient activation-memory peaks and changing compute/bandwidth demands during diffusion inference.
- Co-optimized activation-memory budgeting, phase-aware request scheduling, and sparse attention to translate memory savings into serving efficiency.

APEX: Asynchronous CPU–GPU Execution for LLM Inference Spring 2025

- Built an asynchronous hybrid inference scheduler to overlap CPU-offloaded work with GPU execution during memory-constrained LLM decoding.
- Designed a performance model to guide heterogeneous resource allocation; improved throughput by 84–96% over vLLM on NVIDIA T4 while preserving latency.

SLED: Distributed Speculative LLM Serving Spring 2025

- Contributed to a distributed speculative decoding framework coordinating token drafting on heterogeneous edge devices with batched target-model verification on a shared server.

Publications & Manuscripts

Athanor: An Agent-Native System for Autonomous Post-Training Research
Yufa Zhou, Yichen Wang, **Jiakun Fan**, et al.
arXiv preprint, 2026

Factorization-Guided Training and Inference for Diffusion-Drafted Speculative Decoding
Jiakun Fan, Min Xiao, et al.
Under review

dLLM-Serve: Bridging the Memory Gap in Diffusion Language Model Serving
Jiakun Fan, Yanglin Zhang, Xiangchen Li, Dimitrios S. Nikolopoulos.
ICS 2026

APEX: Asynchronous Parallel CPU-GPU Execution for Online LLM Inference on Constrained GPUs
Jiakun Fan, Yanglin Zhang, Xiangchen Li, Dimitrios S. Nikolopoulos.
IPDPS 2026

WISP: Waste- and Interference-Suppressed Distributed Speculative LLM Serving at the Edge via Dynamic Drafting and SLO-Aware Batching
Xiangchen Li, **Jiakun Fan**, et al.
SIGMETRICS 2026

SLED: A Speculative LLM Decoding Framework for Efficient Edge Serving
Xiangchen Li, Dimitrios Spatharakis, Saeid Ghafouri, **Jiakun Fan**, et al.
ACM/IEEE SEC 2025

Technical Skills

Programming: Python, C/C++, CUDA, Rust
Frameworks & Libraries: PyTorch, vLLM, SGLang, FlashAttention
Systems & Infrastructure: Linux, Docker, GPU experiment orchestration